

Vehicle Architecture-Driven Design and Comprehensive Loss Analysis of an 800V Three-Level Flying Capacitor Traction Inverter

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Abstract—This paper presents the specification, simulation, and loss analysis of a three-level flying capacitor (FC) traction inverter for an 800V electric vehicle powertrain. The 800V DC bus architecture is adopted to reduce phase currents and ohmic losses relative to conventional 400V systems. The FC topology is selected over the Neutral Point Clamped (NPC) alternative on the basis of thermal symmetry and natural voltage balancing under dynamic loads. Conduction and switching losses are quantified using the Graovac-Purschel analytical method, calibrated with parameters from the Wolfspeed CAB450M12XM3 1200V/450A SiC module datasheet. An adaptive switching frequency schedule is proposed that reduces switching losses by 75% during high-current urban operation compared to a fixed 20 kHz baseline. Time-domain simulations confirm successful capacitor voltage balancing, proper pre-charge handling, and safe thermal operation across a representative drive cycle.

I. INTRODUCTION

The traction inverter converts DC battery power into variable-frequency AC power for the electric vehicle (EV) motor. A fundamental constraint in traction inverter design is that the required electrical frequency is directly proportional to vehicle speed: low-speed urban driving demands high phase currents at low fundamental frequencies, while highway cruising requires high frequencies but lower sustained currents. Consequently, evaluating inverter performance requires analysis across the full operating envelope rather than at a single nominal point.

The industry transition to 800V DC bus architectures halves the phase current for a given power level compared to 400V systems, thereby reducing ohmic losses in the inverter, motor windings, and cabling [1]. However, this elevated bus voltage introduces challenges for semiconductor stress management. In a conventional two-level voltage source inverter (2L-VSI), each device must block the full 800V bus, and the fast switching transitions of silicon carbide (SiC) MOSFETs generate high dv/dt rates that degrade motor winding insulation [2]. Multilevel topologies address both issues by synthesizing the output from multiple voltage levels, which reduces the per-device blocking voltage and the voltage step size at the output [3].

To further optimize inverter efficiency, recent literature has explored adaptive or variable switching frequency (VSF) control strategies. Conventional inverters operate at a fixed switching frequency sized for the worst-case thermal condition. In contrast, adaptive strategies dynamically adjust the switching frequency based on the operating point. Prior work has demonstrated the efficacy of VSF for minimizing total drivetrain losses using complex finite element method (FEM) motor models [4] and for reducing DC-link capacitor sizing in GaN-based drives [5]. Other approaches propose varying the

switching frequency within a single fundamental cycle to bound current ripple [6].

Despite these advances, a gap remains in applying adaptive switching frequency specifically to three-level flying capacitor (FC) traction inverters. In the FC topology, the switching frequency cannot be arbitrarily reduced without compromising the natural voltage balancing of the flying capacitors, which requires a sufficient frequency modulation ratio (m_f). Furthermore, while previous studies focus on instantaneous efficiency, the impact of adaptive switching on the thermal cycling amplitude—and by extension, the Coffin-Manson fatigue lifetime of the SiC modules—requires specific quantification.

This paper bridges this gap by proposing a vehicle-architecture-driven adaptive switching frequency schedule for an 800V 3L-FC traction inverter. The main contributions are: 1) a methodology for mapping vehicle kinematics directly to inverter electrical operating points; 2) the formulation of an adaptive frequency schedule that maintains sufficient m_f for capacitor balancing while minimizing losses during high-current urban driving; and 3) the quantification of the resulting thermal cycling reduction using a datasheet-calibrated 4-layer Foster thermal network. The remainder of this paper is organized as follows: Section II details the topology selection, component sizing, and analytical loss formulation. Section III presents the adaptive schedule and time-domain simulation results. Section IV discusses modeling limitations, and Section V concludes the paper.

II. TOPOLOGY AND COMPONENT SELECTION

This section details the topology choice, component specifications, and the vehicle architecture mapping used to define the inverter operating points.

A. Topology Selection: FC vs. NPC

The two principal three-level topologies for traction applications are the Neutral Point Clamped (NPC) and the Flying Capacitor (FC). The FC topology was selected for this design based on three considerations [4].

First, the NPC topology exhibits unequal loss distribution among its inner and outer switches, leading to asymmetric thermal stress that complicates heatsink design [5]. The FC topology distributes losses symmetrically: all four switches in a phase leg block the same voltage ($V_{dc}/2 = 400V$) and carry comparable currents. Second, the FC eliminates the clamping diodes required in the NPC, removing components from the conduction path. Third, the FC topology can maintain capacitor voltage balance through redundant switching states. By

selecting between redundant zero-voltage states based on the instantaneous phase current direction, the flying capacitor voltage is regulated to $V_{dc}/2$ without a dedicated closed-loop controller [6].

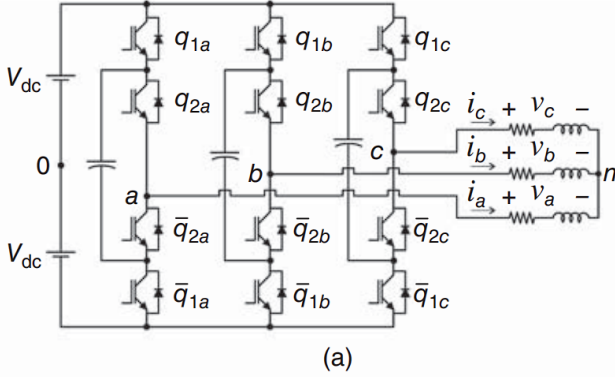


Fig. 1. Three-phase schematic of the 3-level flying capacitor inverter. Source: Advanced Power Electronics Converters (E. Cipriano).

B. Component Selection

The Wolfspeed CAB450M12XM3 SiC half-bridge module was selected for the active switches. In the 3L-FC configuration, each switch blocks $V_{dc}/2 = 400V$. The 1200V device rating therefore provides a voltage utilization of 33%, consistent with IPC-9592B derating guidelines for power conversion equipment [7]. The 450A continuous current rating provides a 3.3× margin over the worst-case peak phase current of 136A.

The total effective on-state resistance per switch position is the sum of the die $R_{ds(on)}$ (2.6 mΩ typical at 25°C) and the package resistance (~0.7 mΩ), yielding approximately 3.3 mΩ at 25°C. From the datasheet normalized $R_{ds(on)}$ curve (Fig. 2 of [8]), this value approximately doubles at 175°C. The positive temperature coefficient is characteristic of SiC MOSFETs and provides inherent current-sharing stability in paralleled configurations.

The flying capacitor must maintain its voltage within a 5% ripple budget (20V on a 400V reference). Using the standard sizing relation $C = I_{pk} / (2 \cdot f_{sw,min} \cdot \Delta V_{fc})$, with $I_{pk} = 136A$, $f_{sw,min} = 5$ kHz, and $\Delta V_{fc} = 20V$, the required capacitance is 680 μF. Metallized polypropylene film capacitors are specified for their low ESR (<5 mΩ), high ripple current capability, and long operational lifetime relative to electrolytic alternatives [9].

C. Vehicle Architecture and Operating Points

The electrical operating points are mapped from the vehicle's mechanical state. The drivetrain parameters are: single-speed reduction gear (ratio $G = 8.19$), tyre radius $r = 0.315$ m, and an 8-pole PMSM (pole pairs $p = 4$). The fundamental electrical frequency is:

$$f_e = (v/r) \cdot (G/2\pi) \cdot p$$

At 40 km/h, $f_e = 184$ Hz; at 120 km/h, $f_e = 552$ Hz. The inverter drives an RL load ($R = 1.3$ Ω, $L = 2$ mH) with modulation index $m = 0.9$. Because the load impedance $Z = \sqrt{R^2 + (\omega L)^2}$ increases with frequency, the peak phase

current is highest at low speeds: 136A at 40 km/h versus 51A at 120 km/h.

D. Loss Formulation

Losses are computed using the analytical method of Graovac and Purschel [10], which averages switching energy over the fundamental cycle. The relevant device parameters from the CAB450M12XM3 datasheet are: $E_{on} = 25.4$ mJ and $E_{off} = 7.51$ mJ at the reference condition of 450A, 600V; $R_{th,jc} = 0.094$ °C/W per switch position [8].

The average switching power loss per phase for the 3L-FC topology is:

$$P_{sw} = N_c \cdot (f_{sw}/\pi) \cdot E_{sw,ref} \cdot (I_{pk}/I_{ref}) \cdot (V_{dc,sw}/V_{ref})$$

where $N_c = 2$ is the number of devices commutating per PWM transition in the 3L-FC leg, $E_{sw,ref} = E_{on} + E_{off} = 32.9$ mJ, $I_{ref} = 450A$, $V_{ref} = 600V$, and $V_{dc,sw} = V_{dc}/2 = 400V$. The $(1/\pi)$ factor arises from averaging the rectified sinusoidal current over one fundamental period. Conduction loss per phase is $P_{cond} = 2 \cdot R_{ds(on)}(T_j) \cdot I_{pk}^2/8$, reflecting that two devices conduct simultaneously and the RMS current per switch is $I_{pk}/2\sqrt{2}$ for sinusoidal modulation.

III. RESULTS

This section presents the analytical loss optimization results and the time-domain simulation validation.

A. Analytical Loss Optimization

Fig. 2 shows the switching loss per phase across the vehicle speed range for several carrier frequencies. At a fixed 20 kHz, switching losses are 84.3 W/phase at 40 km/h (urban) due to the high phase current. Reducing the carrier to 5 kHz at this operating point yields 21.1 W/phase—a 75% reduction.

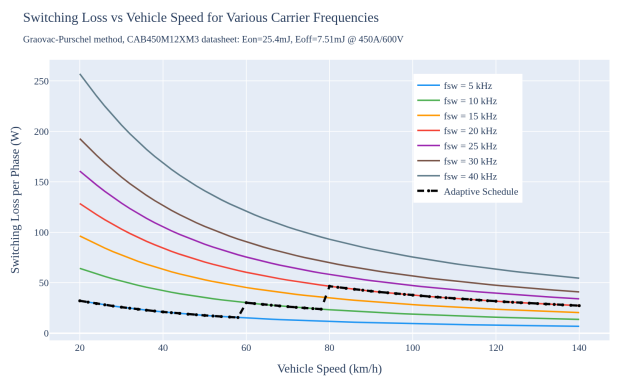


Fig. 2. Switching loss per phase vs. vehicle speed for various carrier frequencies. The dashed line indicates the adaptive schedule.

An adaptive switching frequency schedule was developed to exploit this relationship. The schedule assigns $f_{sw} = 5$ kHz for speeds below 60 km/h (where $f_e < 276$ Hz, giving $m_f \approx 27$), $f_{sw} = 10$ kHz for 60–90 km/h, and $f_{sw} = 20$ kHz above 90 km/h (where $f_e > 414$ Hz, giving $m_f \approx 36$). The constraint is that the frequency modulation ratio $m_f = f_{sw}/f_e$ must remain

sufficiently high to avoid low-order baseband harmonics in the output voltage spectrum.

TABLE I. ANALYTICAL LOSS AND THERMAL COMPARISON

Metric	Urban (40 km/h)	Highway (120 km/h)
Electrical Frequency (f_e)	183.9 Hz	551.7 Hz
Peak Phase Current (I_{pk})	135.8 A	51.0 A
Fixed 20 kHz Switching Loss	84.3 W/phase	31.7 W/phase
Scheduled Switching Loss	21.1 W/phase (5 kHz)	31.7 W/phase (20 kHz)
Steady-State T_j (Fixed 20 kHz)	63.2 °C	46.0 °C
Steady-State T_j (Scheduled)	53.5 °C	46.0 °C

Table I summarizes the comparison at the two primary operating points. The adaptive schedule reduces the three-phase switching loss at 40 km/h by 190 W (from 253 W to 63 W), which corresponds to a 9.7°C reduction in steady-state junction temperature. The thermal analysis uses a 4-layer Foster network parameterized from the CAB450M12XM3 Z_{th} curve ($R_{th,jc} = 0.094$ °C/W total) with a coolant temperature of 65°C.

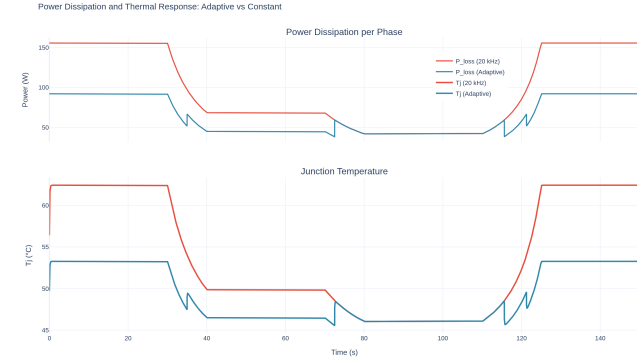


Fig. 3. Power dissipation and junction temperature over a mixed drive cycle comparing the adaptive schedule against a fixed 20 kHz baseline.

A transient thermal simulation over a synthetic drive cycle (Fig. 3) shows that the adaptive schedule reduces the peak-to-peak thermal cycling amplitude from 16.4°C to 7.7°C—a 53% reduction. According to the Coffin-Manson model, this reduction in ΔT_j translates to a substantial extension of the power module's fatigue lifetime, as the number of cycles to failure scales inversely with $(\Delta T_j)^n$ where n is typically 4–5 for wire-bond interconnects [11].

B. Time-Domain Simulation

A time-domain simulation was constructed in Python to validate the design under dynamic conditions. The simulation models the three-phase 3L-FC inverter with Phase-Disposition PWM (PD-PWM), temperature-dependent $R_{ds(on)}$, linear switching energy scaling, a 4-layer Foster thermal network, and dead-time voltage distortion ($t_{dead} = 200$ ns). The time step is

2 μ s and the total simulation spans 1.52 s across nine drive-cycle segments.

Fig. 4 shows the simulated three-phase output currents, which exhibit the expected sinusoidal waveforms with amplitude variation corresponding to the different speed segments. The Total Harmonic Distortion (THD), computed using a Hanning-windowed FFT over 20 fundamental cycles, is 1.92% for the urban segment (5 kHz) and 1.13% for the highway segment (20 kHz). These values represent an optimistic bound, as the simulation does not model device voltage drops in the output voltage synthesis, discrete dead-time distortion, or DC bus voltage ripple—all of which would increase THD in hardware.

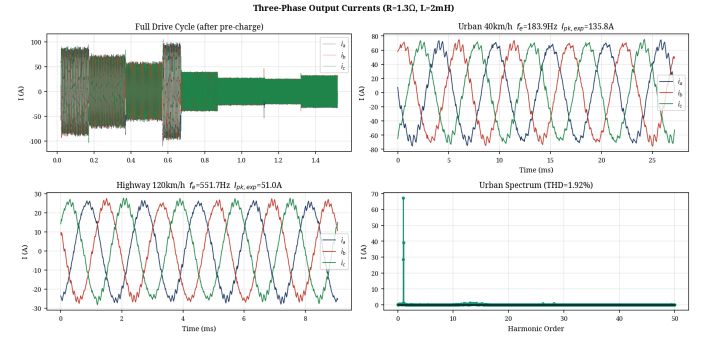


Fig. 4. Simulated three-phase output currents across the multi-segment drive cycle.

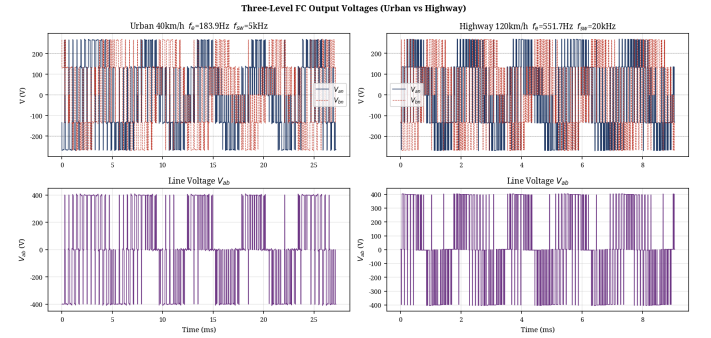


Fig. 5. Three-level output voltage waveforms (line-to-neutral) showing the $\pm V_{dc}/2$ and zero levels.

C. Flying Capacitor Balancing

The simulation incorporates a 20 ms linear pre-charge ramp to bring the flying capacitor voltage from 0V to 400V before active PWM commences, preventing overvoltage across the inner switches at startup. As shown in Fig. 6, the PD-PWM natural balancing algorithm maintains the capacitor voltage with a mean error of less than 0.04V from the 400V reference. The simulated urban ripple is 18.0V peak-to-peak, consistent with the analytical prediction of 20V for the 680 μ F capacitor at 5 kHz and 136A peak current.

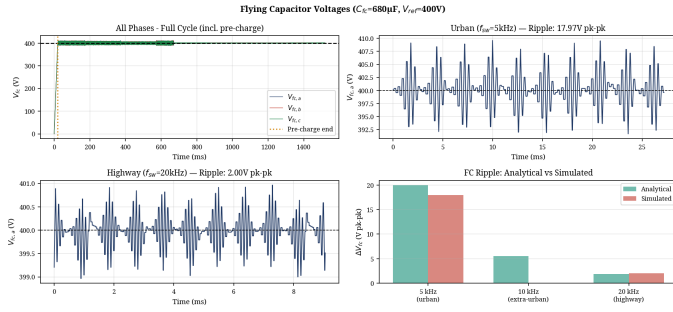


Fig. 6. Flying capacitor voltage showing the 20 ms pre-charge transient and steady-state ripple.

IV. DISCUSSION

Several modeling simplifications should be noted. The simulation uses ideal switching transitions without modeling commutation-loop parasitic inductance (6.7 nH per the datasheet). The dead-time effect is implemented as a continuous average voltage error rather than discrete pulse-by-pulse distortion, which underestimates the resulting low-order harmonics. The DC bus is modeled as an ideal voltage source without ripple. The flying capacitor balancing algorithm updates every carrier period with perfect voltage measurement, which is more frequent and more accurate than would be achievable in a digital controller with ADC noise. These simplifications collectively produce THD values at the optimistic end of the range reported in the literature for 3-level FC inverters: 3–8% at 5 kHz and 1.5–4% at 20 kHz are more typical experimental values [12].

The $R_{ds(on)}$ temperature scaling used in this work ($2\times$ from 25°C to 175°C) is derived from the normalized resistance curve in the datasheet at the rated current of 450A. At the lower operating currents in this design (51–136A), the actual temperature coefficient may differ slightly. This represents a conservative assumption for loss estimation, as it overestimates the resistance at elevated temperatures relative to the tabulated typical value of 4.7 mΩ (die only) at 175°C.

V. CONCLUSION

This work demonstrates that traction inverter design benefits from explicit coupling to the vehicle architecture. The 3L-FC topology was selected over NPC for its symmetric loss distribution and natural capacitor balancing under dynamic loads. Component selection was justified against the

CAB450M12XM3 datasheet and IPC-9592B derating standards. By mapping vehicle speed to electrical operating points, the Graovac-Purschel analytical method quantified a 75% reduction in urban switching losses (84.3 to 21.1 W/phase) when the carrier frequency is reduced from 20 kHz to 5 kHz. Over a representative drive cycle, the adaptive schedule reduces thermal cycling amplitude by 53% (from 16.4°C to 7.7°C), with direct implications for power module fatigue lifetime. Time-domain simulation validated the FC voltage balancing and confirmed THD below 2% across all operating points, though these values represent an optimistic bound due to the ideal switching model employed.

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